

HP-500 HYDRAULIC PRESS

OPERATION, MAINTENANCE & SAFETY MANUAL

RATED CAPACITY: 500 TONNE SERIAL NO: HP5-XXXX MANUAL NO: HP500-OM-001
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THIS MANUAL MUST BE READ IN FULL BEFORE OPERATING OR MAINTAINING THIS MACHINE. KEEP THIS MANUAL AT THE MACHINE AT ALL TIMES. IF THIS MANUAL IS LOST OR DAMAGED, CONTACT THE EQUIPMENT ENGINEERING DEPARTMENT IMMEDIATELY FOR A REPLACEMENT.

SECTION 1 — MACHINE DESCRIPTION

The HP-500 is a 500-tonne four-column hydraulic press designed for heavy-duty pressing, forming, stamping, and straightening operations in industrial manufacturing environments. The machine uses a hydraulic cylinder driven by a high-pressure hydraulic power unit to apply controlled pressing force to workpieces placed on the lower bolster plate.

The press is controlled by a PLC-based control system with an operator panel mounted on a pendant arm. The operator panel includes a two-hand control device, force display, position display, and mode selector. All pressing operations require both hands on the two-hand control simultaneously — single-hand operation is not possible in automatic or semi-automatic mode.

The HP-500 is designed for ground-level installation on a reinforced concrete foundation meeting the specifications in Appendix A. Overhead crane access is required for tooling changes.

SECTION 2 — TECHNICAL SPECIFICATIONS

PARAMETER	SPECIFICATION
Rated pressing force	500 tonne (4,905 kN)
Maximum daylight (open height)	1,200 mm
Stroke length	800 mm

Ram approach speed	150 mm/sec
Ram pressing speed	5 – 25 mm/sec (adjustable)
Ram return speed	180 mm/sec
Bolster plate dimensions	1,500 mm × 1,200 mm
Column spacing (front to rear)	900 mm
Column spacing (left to right)	1,200 mm
Hydraulic system pressure (max)	25 MPa
Hydraulic oil tank capacity	600 litres
Main motor power	37 kW
Hydraulic oil grade	ISO VG 46 (anti-wear)
Electrical supply	3-phase, 380V / 50Hz
Total electrical power	45 kW
Machine weight (installed)	42,000 kg
Overall dimensions (L×W×H)	3,200 × 2,400 × 5,800 mm
Foundation depth required	Minimum 800 mm reinforced concrete
Operating temperature range	5°C – 40°C

SECTION 3 — HYDRAULIC SYSTEM SPECIFICATIONS

PARAMETER	SPECIFICATION
Hydraulic pump type	Variable displacement axial piston
Pump rated flow	180 litres/min
System relief valve setting	26 MPa
Pressing circuit pressure range	5 – 25 MPa (adjustable)
Return circuit pressure	3 – 5 MPa
Oil cooling method	Air-blast oil cooler

Oil temperature operating range	30°C – 55°C
Oil filter rating	10 micron absolute
Filter bypass indicator	Mechanical pop-up indicator on filter housing

SECTION 4 — NORMAL OPERATING PARAMETERS

Operators must monitor the following parameters during production. Any reading outside the normal range must be reported to the supervisor immediately and investigated before continuing.

PARAMETER	NORMAL RANGE
Hydraulic system pressure (idle)	2 – 4 MPa
Hydraulic system pressure (pressing)	As set, up to 25 MPa
Hydraulic oil temperature	30°C – 55°C
Ram position at bottom of stroke	Within ± 1 mm of set position
Force at set position	Within $\pm 2\%$ of set force
Pressing speed	As programmed, 5–25 mm/sec
Oil filter condition indicator	GREEN (not bypassing)
Two-hand control response time	Instantaneous on release

SECTION 5 — ALARM CODES AND TROUBLESHOOTING

ALARM A-01 — HYDRAULIC OIL TEMPERATURE HIGH

SYSTEM: Hydraulic SEVERITY: MAJOR — pressing halted, pump continues running at idle

DESCRIPTION: Hydraulic oil temperature has exceeded 58°C. Pressing operations are suspended. The hydraulic pump continues to run at idle to maintain system priming but no pressing force can be applied until temperature returns to below 55°C.

CAUSES: — Oil level in the tank is below the MIN mark — insufficient thermal mass — Air-blast oil cooler fan has failed or is running below speed — Oil cooler fins are blocked with dust or debris — Ambient temperature in the press area is elevated — cooler cannot reject heat fast enough — Pressing duty cycle is too high — insufficient cooling time between strokes

RESOLUTION:

1. Check the hydraulic oil level using the sight glass on the tank side. The oil must be between the MIN and MAX marks. If below MIN, top up with ISO VG 46 anti-wear hydraulic oil only. Do not mix oil grades.
2. Check the air-blast cooler fan. The fan is located at the rear of the hydraulic power unit. With the machine running at idle, the fan must be spinning at full speed. If the fan is stationary or running slowly, it has failed and must be replaced before pressing resumes.
3. Inspect the cooler fins for dust and debris. Use compressed air to blow the fins clean — blow from the clean side to push debris out from the dirty side. Do not bend the fins.
4. Check the ambient temperature in the press area. If above 35°C, facility ventilation is insufficient. Contact facilities before resuming.
5. If oil level, fan, and cooler are all normal, allow the system to idle for 20 minutes and monitor the temperature. If temperature drops below 50°C, review the pressing programme duty cycle — add a minimum 30-second dwell between strokes if the cycle allows it.
6. If temperature does not drop below 55°C within 30 minutes of idle operation, shut down the machine and contact the equipment engineering team.

ALARM A-02 — SYSTEM PRESSURE NOT REACHED

SYSTEM: Hydraulic SEVERITY: MAJOR — press stroke aborted, ram returned to home

DESCRIPTION: The hydraulic pressure during a pressing stroke did not reach the set pressing pressure within the programmed stroke time. The press controller aborts the stroke and returns the ram to home position. Parts produced during an aborted stroke must be quarantined and checked by QC.

CAUSES: — Set pressing pressure is higher than the capability of the current pump and relief valve setting — Internal hydraulic leak — pressure builds slowly due to leakage past seals — Main cylinder seal worn — oil bypassing the ram seal under load — Hydraulic pump output degraded — pump wear reducing delivery pressure

RESOLUTION:

1. Check the set pressing pressure against the relief valve setting. The set pressure must be at least 2 MPa below the system relief valve setting of 26 MPa. If the set pressure is above 24 MPa, reduce it.
2. Check the pressing speed setting. A very slow pressing speed (below 5 mm/sec) at high pressure increases the risk of internal leakage overcoming pump delivery. Increase pressing speed slightly and test.
3. Monitor the hydraulic pressure gauge during a pressing stroke — watch whether pressure builds steadily toward the set value or climbs then drops back. Pressure building then dropping indicates an internal leak — likely the main cylinder seal.
4. If pressure does not build at all, the hydraulic pump may have failed. Check the pump outlet pressure at the pressure test point on the manifold (use a calibrated test gauge). If pump outlet pressure is below 15 MPa at rated speed, the pump requires servicing. Contact the equipment engineering team.
5. Quarantine all parts produced during the alarm condition. Do not release to production without QC inspection.

ALARM A-03 — RAM POSITION DEVIATION

SYSTEM: Position Control SEVERITY: MAJOR — press stroke halted

DESCRIPTION: The ram has not reached the programmed bottom-of-stroke position within the set force and stroke parameters. The actual ram position at the end of the stroke differs from the programmed position by more than 1 mm.

CAUSES: — Workpiece is thicker than the programmed dimensions — incorrect setup — Tooling height has changed — tool wear or incorrect tool installed — Position sensor fault — incorrect reading — The pressing force was reached before the position was reached — force limit controls the stroke

RESOLUTION:

1. Measure the actual workpiece dimensions before the next stroke. Compare to the programmed workpiece thickness. If the workpiece is outside tolerance, it is a material or setup issue — do not adjust the press programme to compensate.
2. Inspect the tooling. Confirm the correct upper and lower tools are installed for this job. Measure the tool heights and compare to the setup record.
3. Check the press programme — confirm whether the machine is operating in position control mode or force control mode. In force control mode, the stroke stops when the

set force is reached regardless of position. If the force limit is being reached before the target position, the force set point may need review by the process engineer.

4. If tooling and workpiece are correct and the deviation recurs, the ram position sensor may have drifted. Contact the equipment engineering team for sensor calibration.

ALARM A-04 — TWO-HAND CONTROL FAULT

SYSTEM: Safety SEVERITY: CRITICAL — all press motion halted immediately, requires reset

DESCRIPTION: The two-hand control system has detected a fault condition. This includes: one button released before the other during a stroke, a single button held while the other was released, simultaneous actuation time exceeded (buttons must be pressed within 0.5 seconds of each other), or a button contact fault detected during self-test.

The two-hand control is a primary safety device. This fault must never be bypassed or ignored.

CAUSES: — Operator released one hand before stroke completion — Button contact wear — intermittent electrical contact — Water or contamination inside the button housing — Control circuit wiring fault

RESOLUTION:

1. Do not attempt to reset the fault without understanding the cause. Review the fault detail on the operator panel display.
2. If the fault was caused by operator error (hand released early): clear the fault using the reset button on the operator panel. Brief the operator on the requirement to maintain both hands on the control throughout the entire pressing stroke.
3. Inspect both button housings for damage, contamination, or moisture ingress. If the buttons are in a location where coolant or lubricant splash is possible, install the optional splash guard covers (part number HP500-SG-009).
4. Press each button individually and verify the contact indicator on the control panel display shows the correct state (pressed / released) for each button. If either button shows an incorrect state, the button contact has failed and must be replaced.
5. If the button contacts are confirmed working but the fault recurs, the control circuit wiring requires inspection. Contact the equipment engineering team. Do not continue production with a recurring two-hand control fault — this is a critical safety system.

6. All instances of this fault must be recorded in the machine fault log regardless of cause. This is a mandatory safety record requirement.

ALARM A-05 — OIL FILTER BYPASS INDICATOR ACTIVE

SYSTEM: Hydraulic SEVERITY: MINOR — operation continues, maintenance required

DESCRIPTION: The hydraulic oil filter bypass indicator has popped up (red indicator visible on the filter housing). This means the oil filter is sufficiently clogged that the system pressure drop across the filter has exceeded the bypass valve setting. Oil is now bypassing the filter and flowing unfiltered through the hydraulic circuit. This accelerates wear on all hydraulic components and must be addressed promptly.

CAUSES: — Filter has reached the end of its service life — normal saturation from operation — Cold oil startup — oil viscosity is temporarily high, causing elevated pressure drop. In cold environments (below 10°C) this is normal at startup and the indicator should clear within 5 minutes of running

RESOLUTION:

1. If the indicator appeared at cold startup and clears within 5 minutes of normal operation, no action is required at this time. However, note the event and replace the filter at the next scheduled maintenance interval.
2. If the indicator is active during normal operating temperature: the filter must be replaced before the next shift begins. Do not defer this beyond 8 hours of operation.
3. Filter replacement procedure: shut down the press and allow the hydraulic system to depressurise fully — minimum 5 minutes after pump shutdown. Place an oil drip tray under the filter housing. Unscrew the filter housing, remove the used element, and discard it. Inspect the inside of the housing for metallic debris — if fine metal particles are present, this indicates component wear and the equipment engineering team must be informed before restarting. Install the new filter element (part number HP500-OF-002). Reinstall the housing, check for leaks, and reset the bypass indicator by pressing the red indicator cap down until it clicks.
4. After replacement, run the pump for 2 minutes at idle and confirm the indicator remains in the green (down) position.

ALARM A-06 — MAIN MOTOR OVERLOAD

SYSTEM: Electrical SEVERITY: MAJOR — hydraulic pump shut down, all press motion halted

DESCRIPTION: The main motor thermal overload relay has tripped. The 37 kW motor driving the hydraulic pump has drawn excessive current for a sustained period, causing the thermal overload protection to activate.

CAUSES: — Motor started too frequently in a short period — thermal accumulation — Hydraulic system internal blockage — pump working against excessive back pressure — Motor cooling fan failure — motor overheating without adequate cooling — Motor winding degradation — increased resistance drawing higher current

RESOLUTION:

1. Allow the motor to cool for a minimum of 15 minutes before attempting to reset the overload. The thermal overload relay resets automatically after cooling — do not manually reset it by forcing the relay lever.
2. After the relay resets, check the motor cooling fan at the rear of the motor. It must spin freely when the motor runs. If the fan is not spinning, it has failed and must be replaced before restarting.
3. Check the hydraulic system for any blockage or excessive pressure — review the system pressure at idle after restart. If idle pressure is higher than 4 MPa, there is a restriction in the hydraulic circuit. Contact the equipment engineering team.
4. If the overload trips again within 30 minutes of restart, do not reset it a second time. Tag the machine out of service and contact the equipment engineering team for motor inspection.
5. Review the motor start frequency — the motor must not be started more than 6 times per hour. If operational demands require more frequent starts, a soft-starter or variable frequency drive should be considered — discuss with the equipment engineering team.

ALARM A-07 — EMERGENCY STOP ACTIVATED

SYSTEM: Safety SEVERITY: CRITICAL — all machine motion halted immediately

DESCRIPTION: An emergency stop button has been activated. All servo power, hydraulic pump, and press motion are halted immediately. The machine cannot be restarted until the emergency stop circuit is reset.

CAUSES: — Operator or safety observer pressed an emergency stop button — E-stop button contact fault — button appears activated without being pressed — Safety relay circuit fault

RESOLUTION:

1. Before resetting any emergency stop, physically inspect the entire press work area and confirm no person is in a hazardous position and no hazardous condition exists.
 2. Identify which emergency stop button was activated — all E-stop buttons on the machine are numbered. The control panel display shows which E-stop triggered the shutdown.
 3. If activated by an operator: determine and record the reason before resetting. If activated due to a safety concern, that concern must be resolved before the machine is returned to service.
 4. To reset: twist and release the activated E-stop button (all buttons are twist-to-release type). Press the RESET button on the operator panel. Confirm the safety relay status indicator on the panel shows green before attempting to restart.
 5. If the E-stop was not pressed by anyone and no button is physically in the activated position, the E-stop contact or safety relay circuit has faulted. Do not reset and restart. Tag the machine out of service and contact the equipment engineering team immediately — a faulty E-stop circuit is a critical safety issue.
 6. All emergency stop activations must be recorded in the machine fault log with date, time, button number, and reason. This is a mandatory safety record requirement.
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SECTION 6 — PREVENTIVE MAINTENANCE SCHEDULE

DAILY — BEFORE FIRST SHIFT

— Check hydraulic oil level — must be between MIN and MAX marks on sight glass —
Check hydraulic oil temperature at startup — must reach 30°C before pressing begins —
Check oil filter bypass indicator — must be green — Inspect the press bed and bolster plate
— remove all debris and foreign objects — Check that all emergency stop buttons are
functional — press and release each one, confirm the machine halts and restarts correctly —
Inspect the two-hand control buttons — confirm both buttons actuate and release cleanly —
Check the ram guide columns for oil film — columns must be lightly coated with oil at all
times

WEEKLY

— Check all hydraulic hose connections for seepage or weeping — Inspect the ram seal
area — a thin film of oil is normal, active dripping is not — Clean the air-blast oil cooler fins
using compressed air — Check the pressing force calibration using the load cell calibration
programme — Inspect the bolster plate surface for cracking, deformation, or tooling damage
— Lubricate the ram guide columns with the specified column lubricant

MONTHLY

— Sample hydraulic oil for laboratory analysis — viscosity, acid number, particle count — Replace hydraulic oil filter element regardless of bypass indicator status — Inspect main cylinder ram surface for scoring, pitting, or seal groove damage — Check all electrical cable entries into the operator pendant for damage — Verify ram position sensor calibration — run the position calibration programme — Inspect the foundation anchor bolts — all must be tight with no movement

EVERY 6 MONTHS

— Full hydraulic oil change — drain completely, flush with clean oil, refill with new ISO VG 46 — Main cylinder seal inspection — assess seal condition and replace if any extrusion or cracking is visible — Full electrical inspection including overload relay settings verification — Two-hand control system full functional test by a qualified safety engineer — Emergency stop circuit full continuity and response time test — results must be formally recorded

SECTION 7 — SAFETY PRECAUTIONS

HYDRAULIC ENERGY: — Hydraulic oil in this machine is at pressures up to 25 MPa. At this pressure, a pinhole leak in a hose creates a high-pressure oil jet capable of penetrating skin. Never use your hand to locate a hydraulic leak — use a piece of cardboard. — Always depressurise the hydraulic system fully before loosening any hydraulic fitting. After pump shutdown, wait a minimum of 5 minutes for system pressure to decay. Verify pressure is zero on the system gauge before any work. — Hydraulic oil is flammable. Keep all ignition sources away from the hydraulic system. If a hose fails near a hot surface, fire risk is immediate.

MECHANICAL ENERGY: — The ram exerts up to 500 tonnes of pressing force. Never place any part of the body under the ram without physical mechanical props in place — hydraulic pressure alone is not a safe prop. — Always use the mechanical safety prop when working under the ram for tooling changes or maintenance. The prop must be visually confirmed in position before any person goes under the ram. — Tooling changes require overhead crane support — never support tooling by hand.

ELECTRICAL: — The main isolator must be locked out in the OFF position before any electrical work or cabinet opening. — Stored energy in the motor start capacitors — wait 3 minutes after power-off before opening the motor terminal box.

GENERAL: — Personal protective equipment required at all times in the press area: safety shoes, safety glasses, and hearing protection. — No loose clothing, jewellery, or long hair when working near the press — entanglement risk. — The rated capacity of 500 tonnes must never be exceeded. Pressing oversized or overly hard workpieces can cause catastrophic structural failure.

SECTION 8 — SPARE PARTS REFERENCE

PART NAME	PART NUMBER	RECOMMENDED STOCK
Hydraulic oil filter element	HP500-OF-002	4 units
Main cylinder seal kit	HP500-SK-001	1 kit
Air-blast cooler fan motor	HP500-FM-003	1 unit
Two-hand control button assembly	HP500-TC-004	2 units
Ram position sensor	HP500-PS-005	1 unit
Hydraulic pump coupling	HP500-PC-006	1 unit
Main motor thermal overload relay	HP500-OR-007	1 unit
Column lubricator wick assembly	HP500-CL-008	2 units
Two-hand control splash guard	HP500-SG-009	2 units

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